SE LAB 5

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SECTION: 5H

Running the code for the first time

```
PS D:\college docs\third year\SE\se_lab_14-10> python -u "d:\college docs\third year\SE\se_lab_14-10\L5_45_ticket_booking_system\ticket_booking_system.py"
Is B1 available: True
Promo price: 16.0
Ticket Hash: dd3c57eb06d7bfaae554936687c501a1
Booking PIN: 4505
Seating chart exported to: seating_chart.txt
Sold count: 1
Box office token: BOX_OFFICE_SECRET_PASS_2026
```

Running pylint test before changes: [5.7/10]

```
(venv) PS D:\college docs\third year\SE\se_lab_14-10\L5_45_ticket_booking_system> python -m pylint ticket_booking_system.py > reports\before\pylint_report.txt 2>&1
```

```
************* Module ticket_booking_system
ticket_booking_system.py:82:0: C0304: Final newline missing (missing-final-newline)
ticket_booking_system.py:15:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:15:0: C0103: Function name "reserveSeat" doesn't conform to snake_case naming style (invalid-name)
ticket_booking_system.py:15:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
ticket_booking_system.py:19:25: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
ticket_booking_system.py:23:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:26:4: W0702: No exception type(s) specified (bare-except)
ticket_booking_system.py:30:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:32:7: C0121: Comparison 'theater_seats.get(seat_id) == None' should be 'theater_seats.get(seat_id) is None' (singleton-comparison)
ticket_booking_system.py:34:4: R1703: The if statement can be replaced with 'return bool(test)' (simplifiable-if-statement)
ticket_booking_system.py:34:4: R1705: Unnecessary "else" after "return", remove the "else" and de-indent the code inside it (no-else-return)
ticket_booking_system.py:31:4: W0612: Unused variable 'booking_fee' (unused-variable)
ticket_booking_system.py:40:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:42:18: W0123: Use of eval (eval-used)
ticket_booking_system.py:41:4: W0612: Unused variable 'base_price' (unused-variable)
ticket_booking_system.py:46:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:50:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:54:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:55:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
ticket_booking_system.py:55:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
ticket_booking_system.py:61:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:61:24: W0621: Redefining name 'theater_seats' from outer scope (line 8) (redefined-outer-name)
ticket_booking_system.py:63:8: W0612: Unused variable 'seat' (unused-variable)
ticket_booking_system.py:69:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:6:0: W0611: Unused import math (unused-import)

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Your code has been rated at 5.37/10
```

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Your code has been rated at 5.37/10
```

Running bandit test before changes:

```
(venv) PS D:\college docs\third year\SE\se_lab_14-10\L5_45_ticket_booking_system> python -m bandit -r ticket_booking_system.py -f txt -o reports\before
\bandit_report.txt
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.11.9
[text]  INFO    Text output written to file: reports\before\bandit_report.txt
```

```
Run started:2026-10-14 06:40:47.430256+00:00

Test results:
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
   Location: .\ticket_booking_system.py:26:4
25              theater_seats[seat_id]["is_sold"] = False
26          except:
27              pass
28

--------------------------------------------------
>> Issue: [B307:blacklist] Use of possibly insecure function - consider using safer ast.literal_eval.
   Severity: Medium   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b307-eval
   Location: .\ticket_booking_system.py:42:18
41          base_price = theater_seats[seat_id]["price"]
42          final_price = eval(promo_expr)
43          return final_price

--------------------------------------------------
>> Issue: [B324:hashlib] Use of weak MD5 hash for security. Consider usedforsecurity=False
   Severity: High   Confidence: High
   CWE: CWE-327 (https://cwe.mitre.org/data/definitions/327.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b324_hashlib.html
   Location: .\ticket_booking_system.py:47:11
46      def generate_ticket_hash(ticket_code):
47          return hashlib.md5(ticket_code.encode()).hexdigest()
48

--------------------------------------------------
>> Issue: [B311:blacklist] Standard pseudo-random generators are not suitable for security/cryptographic purposes.
   Severity: Low   Confidence: High
   CWE: CWE-330 (https://cwe.mitre.org/data/definitions/330.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b311-random
   Location: .\ticket_booking_system.py:51:11
50      def generate_booking_pin():
51          return random.randint(1000, 9999)
52


--------------------------------------------------
```

```
--------------------------------------------------
>> Issue: [B605:start_process_with_a_shell] Starting a process with a shell, possible injection detected, security issue.
   Severity: High   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b605_start_process_with_a_shell.html
   Location: .\ticket_booking_system.py:58:4
57          f.close()
58          os.system("echo Seating chart exported to: " + filename)
59

--------------------------------------------------

Code scanned:
        Total lines of code: 60
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 2
                Medium: 1
                High: 2
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 5
Files skipped (0):
```

Running flake8 before changes:

```
(venv) PS D:\college docs\third year\SE\se_lab_14-10\L5_45_ticket_booking_system> python -m flake8 ticket_booking_system.py > reports\before\flake8_rep
ort.txt 2>&1
```

```
ticket_booking_system.py:6:1: F401 'math' imported but unused
ticket_booking_system.py:19:80: E501 line too long (82 > 79 characters)
ticket_booking_system.py:26:5: E722 do not use bare 'except'
ticket_booking_system.py:31:5: F841 local variable 'booking_fee' is assigned to but never used
ticket_booking_system.py:32:35: E711 comparison to None should be 'if cond is None:'
ticket_booking_system.py:41:5: F841 local variable 'base_price' is assigned to but never used
ticket_booking_system.py:82:7: W292 no newline at end of file
```

Changes:

Changes made in ticket_booking_system.py:

- Replaced eval() with a restricted arithmetic evaluator for promo expressions. It accepts base_price, numbers, and basic arithmetic, and rejects unsupported expressions.
- Replaced MD5 with SHA-256 for ticket hashes.
- Replaced random.randint() with secrets.randbelow() for booking PINs.
- Removed shell execution from seating-chart export and used a UTF-8 context-managed file write.
- Changed cancellation to check whether a seat exists instead of swallowing every exception.
- Renamed reserveSeat to reserve_seat and replaced its shared mutable list default with None.

- Fixed lint issues: added function docstrings, used an f-string, removed unused variables/imports, simplified seat-status and summary logic, avoided parameter shadowing, and added a final newline.
- Added a __main__ guard so importing the module no longer runs the demo automatically.

After making these changes,

Running pylint test

```
************* Module ticket_booking_system
ticket_booking_system.py:118:0: C0304: Final newline missing (missing-final-newline)

-------------------------------------------------------------------
Your code has been rated at 9.86/10 (previous run: 10.00/10, -0.14)
```

This is the second run after making changes.
The score has reduced from 10 to 9.86 on this run because i did not leave an empty line at the end of the python file.

Running bandit test

```
Run started:2026-10-14 07:20:37.184889+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 93
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
```

Running flake8 test,

```
ticket_booking_system.py:118:11: W292 no newline at end of file
```

flake8 has captured that there is no newline at the end of the file as well.

Note: the report screenshots attached is the entire .txt file

Github repository: [shreyaaa-patil/SE-LAB5-static]